

values

4. Find the average value of each pixel

5. Create the output file and modify the header suitably

In a BMP file, the first 54 bytes form the header. In a 24-bit BMP file, the image data begins soon after the header. The values of each pixel can be read in the order of blue, green and red. Each of these can have a value between 0 and 255. The average value is found as $(R+G+B)/3$. The process is repeated for all the pixels. This forms the new image data.

To create the output file, first the 54-byte header is copied from the input. The value of the number of bits per pixel located at offset 28 in the header is changed to 8. Then the

palette is created and appended to the output file. Each entry in the palette consists of four values. Of these, three values correspond to blue, green and red, respectively, and the fourth is always zero. 256 entries are made in the palette. Now the image data obtained by taking the average of colour information is appended to the output file. The value of offset to the start of the image (10 in the header) is also replaced with a new value. Now the output file is ready and can be viewed using any image viewer software.

Testing

Compile the source code using a C++ compiler and create an executable file. Place the executable file in the same

folder as the input file. Run the executable file. Enter the input and output file names. Make sure that both the file names are different. Also, the file name should have less than eight characters and the file extension should not be specified. The output file will be created in the same folder.

Most digital cameras store photos in JPEG format. To convert it into BMP, first open it in MS Paint. Then go to File menu and click 'Save As' option. Change 'Save As Type' option to 24-bit bitmap and click 'Ok' button.

EFY note. The source code of this article is available on www.efymag.com website.

The author is a B.Tech in electronics & communications and a hobbyist designer of electronic circuits, software and websites

IMAGE.CPP

```
#include<iostream.h>
#include<conio.h>
#include<fstream.h>
#include<string.h>
#include<stdio.h>

char filename1[25], filename2[25];

int rgbtorgrey()
{
    char c;
    int m,n;
    ifstream fin(filename1, ios::in | ios::binary);
    if (!fin)
        return 0;
    ofstream fout(filename2, ios::out | ios::binary);
    for (int i=0; i<54; i++) //Copy header from input file to output file
    {
        fin.read((char *)c, sizeof(c));
        fout.write((char *)c, sizeof(c));
    }
    for (int il=0; il<256; il++) //Create palette in output file
    {
        c=il;
        fout.write((char *)c, sizeof(c));
        fout.write((char *)c, sizeof(c));
        fout.write((char *)c, sizeof(c));
        c=0;
        fout.write((char *)c, sizeof(c));
    }
    while (!fin.eof()) //Read the three values and find average
    {
        m=0;
        fin.read((char *)c, sizeof(c));
        n=c;
        if (n<0)
            n+=256;
        m+=n;
        fin.read((char *)c, sizeof(c));
        n=c;
        if (n<0)
            n+=256;
        m+=n;
        fin.read((char *)c, sizeof(c));
        n=c;
        if (n<0)
            n+=256;
        m+=n;
        if (m>255)
            m=255;
        else if (m<0)
            m=0;
        c=m;
        fout.write((char *)c, sizeof(c)); //Write image data to output file
    }
    fout.seekp(28); //Number of bits per pixel
    c=8;
    fout.write((char *)c, sizeof(c));

    fout.seekp(11); //Offset to start of image data
    c=4;
    fout.write((char *)c, sizeof(c));
    fout.close();
    fin.close();
    return 1;
}

void main()
{
    clrscr();
    int r;
    cout<<"\nConvert 24 Bit bmp file to 8 Bit bmp file\n\n";
    cout<<"\nNo need to specify .bmp file extension along with file name\n";
    cout<<"\nEnter source file name : ";
    gets(filename1);
    cout<<"\nEnter required output file name : ";
    gets(filename2);
    strcat(filename1, ".bmp");
    strcat(filename2, ".bmp");
    r = rgbtorgrey();
    if (r==1)
        cout<<"\n\nConversion successful..!!";
    else
        cout<<"\n\nError - File not found..!";
    getch();
}
```

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